



Islamic University of Technology
Department of Computer Science and Engineering

Lab 5: Shortest Path (Continued)

CSE 4404: Algorithms Lab(SWE)
Summer 2023-24

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Task A. Crisis Alert

Time Limit: 2 seconds | Memory Limit: 512 MB

A city has n locations connected by m one-way roads. K of these locations have emergency broadcast towers. When an alert is triggered, all towers broadcast simultaneously. For each location, find the minimum time for the alert to arrive from **any** tower.

Input Format

The first line contains three integers n , m , and K ($1 \leq n \leq 2500$, $1 \leq m \leq 5000$, $1 \leq K \leq n$) — the number of locations, roads, and towers.

Each of the next m lines contains three integers a , b , and c ($1 \leq a, b \leq n$, $-10^9 \leq c \leq 10^9$) — a one-way road from location a to location b with travel time c . It is guaranteed that the graph contains **no negative cycles**.

The next line contains K integers — the locations of the broadcast towers.

Output Format

Print n lines. The i -th line contains the minimum travel time from any tower to location i .

If location i is unreachable from all towers, print INF.

Examples

Sample Input	Sample Output
6 6 2	0
1 2 -1	-1
1 3 8	1
2 3 2	0
4 3 1	5
4 5 5	INF
3 5 4	
1 4	

Notes

In the sample, towers are at locations 1 and 4:

- Location 1: already a tower, time = 0.
- Location 2: only reachable from tower 1 via $1 \rightarrow 2$, time = -1.
- Location 3: tower 1 via $1 \rightarrow 2 \rightarrow 3$ gives $-1 + 2 = 1$; tower 4 via $4 \rightarrow 3$ gives 1. Minimum = 1.
- Location 4: already a tower, time = 0.
- Location 5: tower 1 via $1 \rightarrow 2 \rightarrow 3 \rightarrow 5$ gives $1 + 4 = 5$; tower 4 via $4 \rightarrow 5$ gives 5. Minimum = 5.
- Location 6: unreachable from all towers, print INF.

Task B. The Signal Leak

Time Limit: 2 seconds | Memory Limit: 512 MB

You are given a directed weighted graph of n nodes and m edges. Edge weights can be negative. Starting from source node s , find the minimum cost to reach every other node.

Some paths may pass through a **negative cycle** — a cycle whose total weight is negative. Any node reachable via such a cycle has an unbounded negative cost ($-\infty$).

Input Format

The first line contains three integers n , m , and s ($1 \leq n \leq 2500$, $1 \leq m \leq 5000$, $1 \leq s \leq n$) — the number of nodes, edges, and the source node.

Each of the next m lines contains three integers a , b , and c ($1 \leq a, b \leq n$, $-10^9 \leq c \leq 10^9$) — a directed edge from node a to node b with weight c .

Output Format

Print n lines. The i -th line contains the minimum cost from s to node i :

- If $i = s$, print 0.
- If node i is reachable via a negative cycle, print $-\text{INF}$.
- If node i is unreachable from s , print INF .
- Otherwise, print the minimum cost.

Examples

Sample Input	Sample Output
6 5 1 1 2 1 2 3 2 3 2 -5 3 4 1 1 5 8	0 -INF -INF -INF 8 INF
5 6 1 1 2 1 2 3 2 3 4 1 4 2 -10 1 5 2 4 5 1000000000	0 -INF -INF -INF -INF -INF

Notes

In the sample, starting from node 1:

- Edges $2 \rightarrow 3 \rightarrow 2$ form a negative cycle with total weight $2 + (-5) = -3 < 0$. Nodes 2 and 3 are in the cycle.
- Node 4 is downstream of the cycle (reachable from node 3), so it is also $-\text{INF}$.
- Node 5 is only reachable via $1 \rightarrow 5$ with cost 8, unaffected by any cycle.
- Node 6 has no incoming edges so it is unreachable from node 1: INF .

Task C. Dekisugi’s Childhood Dream

Time Limit: 2 seconds | Memory Limit: 512 MB

Hidetoshi Dekisugi was one of the brightest students in the country. He grew into a renowned astronaut and scientist. After a series of successful space missions, he is now working to fulfill a childhood dream. Ever since he saw Doraemon, he secretly wished to build a real-life time machine.

While progress has been slow, Dekisugi recently made a promising discovery — certain “wormholes” scattered across the galaxy. Each wormhole is a one-way portal that instantly transfers someone from one star system to another, but with a twist: it also shifts them forward or backward in time. The time difference can be positive or negative.

Dekisugi believes that if there exists a sequence of wormholes such that a traveler can return to the original star system and go back in time overall, this could form the core mechanism of a time machine.

You are given a map of these wormholes. Can you help Dekisugi determine whether such a **time loop** (a cycle with negative total time shift) exists?

Input Format

The first line contains two integers n and m ($1 \leq n \leq 2500$, $1 \leq m \leq 5000$) — the number of star systems and the number of wormholes.

Each of the next m lines contains three integers a , b , and c ($1 \leq a, b \leq n$, $-10^9 \leq c \leq 10^9$) — indicating that there is a one-way wormhole from star system a to star system b that shifts the traveler by c units of time.

Output Format

Print YES if there exists any time loop (a cycle with total negative time shift). Otherwise, print NO.

Examples

Sample Input	Sample Output
4 5 1 2 3 2 3 2 3 4 -1 4 2 -2 1 3 -5	YES
5 5 3 2 2 3 1 1 2 4 -4 4 3 1 4 5 1	YES

Notes

In sample 1, the wormhole sequence $2 \rightarrow 3 \rightarrow 4 \rightarrow 2$ forms a cycle with total time shift $2 + (-1) + (-2) = -1$, which is negative. So the correct answer is YES.

In sample 2, the wormhole sequence $3 \rightarrow 2 \rightarrow 4 \rightarrow 3$ forms a cycle with total time shift $2 + (-4) + 1 = -1$, which is negative. So the correct answer is YES.

Marks Distribution

Task	Marks
Task A	30%
Task B	40%
Task C	30%

Practice Problems

Problem	Links
Signal Hill	DMOJ: https://dmoj.ca/problem/dmpg15s4 Vjudge: https://vjudge.net/problem/DMOJ-dmpg15s4
Roads in Berland	Codeforces: https://codeforces.com/contest/25/problem/C Vjudge: https://vjudge.net/problem/CodeForces-25C
High Score	CSES: https://cses.fi/problemset/task/1673 Vjudge: https://vjudge.net/problem/CSES-1673
Cycle Finding	CSES: https://cses.fi/problemset/task/1197 Vjudge: https://vjudge.net/problem/CSES-1197
Edge Deletion	AtCoder: https://atcoder.jp/contests/abc243/tasks/abc243_e Vjudge: https://vjudge.net/problem/AtCoder-abc243_e
Flight Discount	CSES: https://cses.fi/problemset/task/1195 Vjudge: https://vjudge.net/problem/CSES-1195
Greg and Graph	Codeforces: https://codeforces.com/problemset/problem/295/B Vjudge: https://vjudge.net/problem/CodeForces-295B